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A positive conglomerate test for the characteristic remanent magnetization (ChRM) of the Emeishan basalt from Southwest China

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Abstract. A positive conglomerate test has been achieved on ten basalt clasts from a pyroclastic breccia in the Emeishan basaltic sequence from Southwest China. Nine basalt samples collected from a nearby lava flow yield ChRM directions similar to the northeasterly and shallow up directions reported from elsewhere in the basalt province. This effectively refutes the suggestion that this group of directions represents a remagnetization of as late as Jurassic or Triassic age. The difference between different sets of results from the Emeishan Basalt and the non-antipodality between the northeasterly directions and the reversed southwesterly down directions are likely to be due to a combination of minor structural rotations about vertical axes between the sampled localities, inaccuracy in determination of bedding attitude in massive basaltic rocks, and insufficient paleomagnetic sampling.

Introduction

McElhinny and his collaborators (1981) conducted a pioneering paleomagnetic investigation on the Late Permian Emeishan Basalt from a type section in Southwest China that was followed by many later studies (Lin, 1984; Huang et al., 1986; Liu et al., 1987; Zhou et al., 1988; Zhao and Coe, 1989; Van der Voo et al., 1993). However, uncertainty still lingers as to whether the observed characteristic direction is primary or secondary. The lingering doubts of remagnetization mainly originate from differences of mean directions from different localities and lack of antipodality between the northeasterly and shallow up directions and the reversed southwesterly down directions. Although a positive fold test has been achieved for the northeasterly directions in some of the studies (McElhinny et al., 1981; Van der Voo et al., 1993), it only demonstrates at best that the acquisition of the magnetization predates the earliest Tertiary, which is the generally accepted age of folding that is much younger than the age of the rocks. A field test to better constrain the age of the ChRM has thus become very much needed. To this end, we recently conducted a conglomerate test on pyroclastic breccia from the basaltic sequence.

Sampling

The samples were taken from northwestern Guizhou Province, which is part of the Emeishan basalt province, a typical continental flood basalt covering a large area of the southwestern Yangtze Block (Fig. 1). The basaltic sequence in western Guizhou Province is characterized by the presence of abundant agglomerate and pyroclastic breccia in the lower part, and a predominance of massive basaltic rocks such as aphyric, porphyritic and amygdaloidal basalts in the middle part, and the occurrence of intercalations of tuff and tuffite in the upper part (Zheng et al., 1978). This apparently reflects a process of initial explosive volcanicity followed by vigorous outpouring of lava flows and ending with intermittent eruptions of waning intensity. We collected samples at two close localities in Weining County at the eastern limb of a NNE-SSW trending syncline (Fig. 1). The basaltic sequence there is 1250 m thick and twenty-two flows are recognized by local geologists (Zheng et al., 1978). The age of the sequence is well constrained as Late Permian (Longtanian) since it disconformably overlies the Late Permian Maokou limestone and disconformably underlies the Xuanwei Formation, a terrestrial coal-bearing clastic sequence of latest Permian (Changxingian) age (Yang et al., 1986; Harland et al., 1990). At the first locality, Heishitou (26.7°N, 104.0°E), a township 30 km southwest of downtown Weining, the rocks dip southwest at 30°-

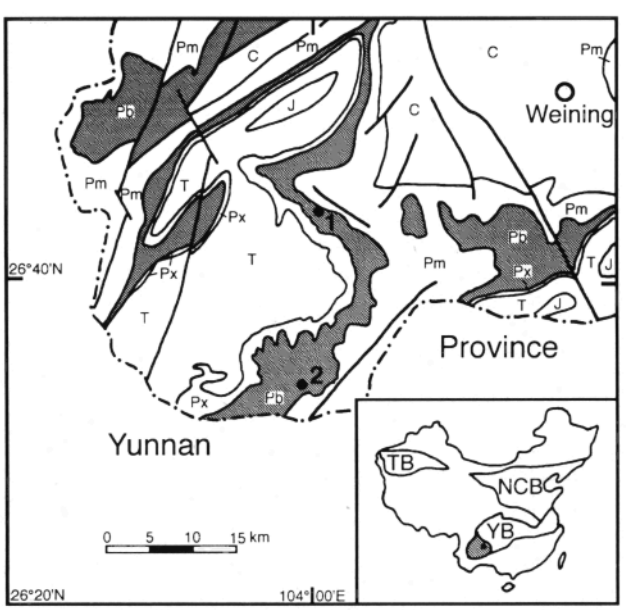


Figure 1. Sketch geologic map of the sampling area. Thick lines are faults, thin lines are boundaries between rock units, and the dot-dash line is the administrative boundary between Guizhou and Yunnan province. Numbered dots denote two sampling localities in this study: 1. Heishitou, and 2. Jule. Rock units: J = Jurassic; T = Triassic; Px = Xuanwei Fm; Pb = Emeishan Basalt (shaded); Pm = Maokou Fm; and C = Carboniferous. The inset shows three major cratons in China (NCB = North China Block; YB = Yangtze Block; and TB = Tarim Block) and the Emeishan Basalt province (shaded) in the southwestern Yangtze Block.

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33°, and nine cores were drilled from an outcrop of dark grey aphyric basalt about 15 m in thickness in the basal part of the basaltic sequence using a portable gasoline-powered drill. In a deeply dissected ravine 20 km south of Heishitou near a village called Jule (26.5°N, 104.0°E), thirteen cores were drilled from the basalt clasts in a pyroclastic breccia bed about 10 m thick in the lower part of the basaltic sequence. The pyroclastic breccia is greenish grey in color and is composed principally of angular basalt clasts with subordinate limestone clasts cemented by basalt. Limestone clasts were not sampled. The vertical relationship between these two sites is not precisely determined, but the sampled basalt flow at Heishitou is likely stratigraphically below the pyroclastic breccia sampled at Jule. It was difficult to recognize bedding in the basaltic rocks at the second site, so a bedding attitude of dipping NE 64.5° at 16° was taken from the underlying limestone for tilt correction for the clast samples. This bedding attitude is a bit disconcordant with the general dip direction of the fold limb (see Fig. 1); nevertheless, it will not affect the grouping of the magnetic directions of the clasts from this bed. Core orientation and bedding attitude were measured using a magnetic compass. The effect of the magnetization of the basaltic rocks on the compass was checked by gradually reducing the distance from the compass to the rock surface, and no significant deflection was detected.

Paleomagnetic results

A specimen one centimeter in length was prepared from each core in the laboratory. Progressive thermal demagnetization was employed for magnetic cleaning and was carried out in a TSD-1 Schonstedt thermal demagnetizer. Magnetization of the samples was measured in a 2G cryogenic magnetometer. All the instruments are housed in a magnetically shielded room. Magnetic directions of individual samples were determined utilizing principal component analysis technique (Kirschvink, 1980) and site mean directions were computed with Fisher (1953) statistics.

Intensities of natural remanent magnetization (NRM) of the basalt samples from Heishitou range from 0.7 to 12 A/m. During thermal demagnetization, a low temperature component (LTC) was removed by 450°C (Figs. 2a and 2b). This LTC only accounts for 5-11% of the total NRM in the very strongly magnetized samples with initial NRM intensities ranging from 4.8 to 12 A/m (Fig. 2a), but it contributes 34-66% to the total NRM intensities in the less strongly magnetized samples (Fig. 2b). The LTC is occasionally aligned with the present-day Earth's magnetic field (Fig. 2b). Above 450°C, a ChRM component was isolated from all nine samples and unblocked at around 570°C, indicating that magnetite is the magnetic carrier. The ChRM directions of the samples are distinct from the present geomagnetic direction at the sampling site before tilt correction, and are northerly with very shallow, mostly upward, inclinations after correction for bedding tilt (Fig. 3 upper). A mean direction of D_g/I_g = 351.9°/-28.4°, D_s/I_s = 359.3°/-5.9° with α_{95} = 10.0°, κ = 27.6 , n = 9 was computed for this site, which yields a VGP at 60.3°N/285.4°E with dp = 5.0° and dm = 10.0°.

Of the thirteen basalt clast samples drilled from the pyroclastic breccia at Jule, ten have NRM intensities comparable to those of the basalt samples from Heishitou and range from 0.2 to 5.4 A/m. Likewise, they possess two components: a LTC was eliminated below 450°C in most cases and a ChRM component was resolved above 500°C (Figs. 2d, 2e and 2f). In most of these samples, the LTC accounts for an appreciable proportion (29-88%) of the total NRM intensities. The remaining three clast samples are much more weakly magnetized with initial NRM intensities ranging from 0.3 to 2.7 mA/m. Inspection of thin sections from these three samples has revealed that two of them are altered basaltic breccia with large amounts of secondary calcite and the third one is a badly altered basalt in which plagioclase and pyroxene are replaced by sericite and chlorite. Among these three samples, only one basaltic breccia sample yielded a ChRM direction (Fig. 2c). Because of their severe alteration nature and resultant weak magnetization, these three samples were excluded from further analysis. The ChRM directions of the ten basalt clasts are quite distinct from one another, and this can be easily seen from their stereographic projections (Fig. 3 lower). On

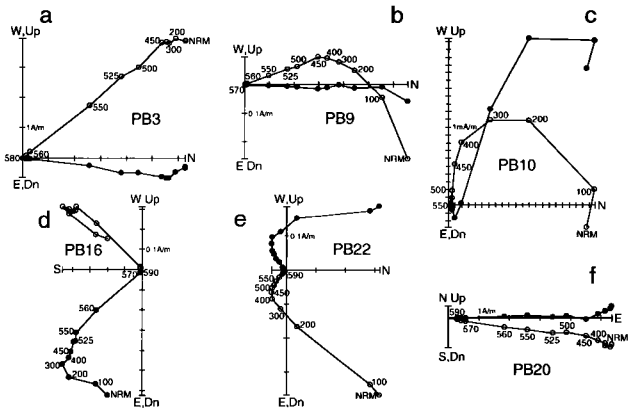


Figure 2. Representative orthogonal plots of basalt (a and b) and clast (c, d, e and f) samples in geographic coordinates. Solid (open) circles are plotted on horizontal (vertical) planes. Thermal treatment levels are in degrees Celsius.

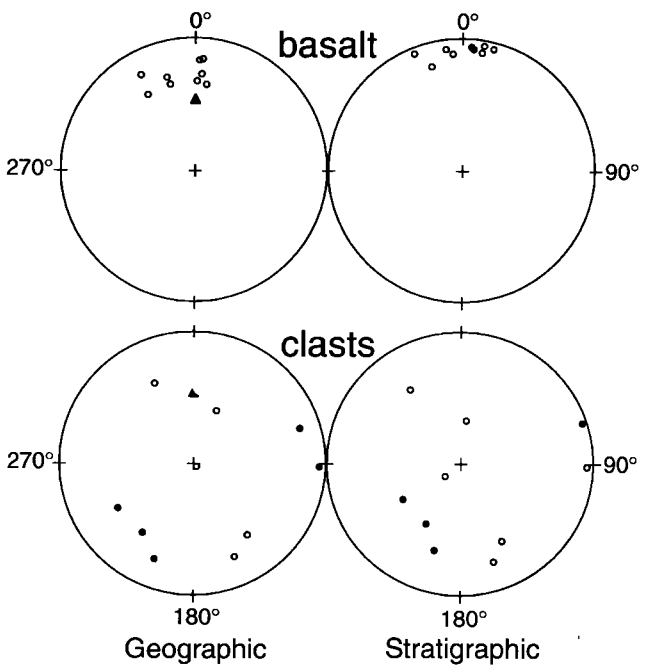


Figure 3. Equal-area projections of the ChRM directions of the basalt (upper) and basalt clast (lower) samples. Solid (open) circles represent directions plotted onto the lower (upper) hemisphere. Black triangle denotes the present dipole field direction at the sampling locality.

Table 1. Paleomagnetic Results from the Emeishan Basalt

Nr.	Localities	Coordinates	D_S/I_S	N/n	α_{95}	Paleopole	A95	P	Reference
1	Emei	29.6°N, 103.4°E	17.8°/ -6.9°	20/72¶	6.4°	52.8°N, 253.1°E	4.5°*	N	McElhinny et al. (1981)
2	Emei	29.6°N, 103.4°E	19.6°/-11.6°	/44	3.6°	49.7°N, 252.0°E	2.8°	N	Zhao and Coe (1989)
3	Emei	29.7°N, 103.5°E	38.4°/-11.8°	/46	5.1°	38.7°N, 231.1°E	3.7°*	N	Liu et al. (1987)
4	Huidong	26.7°N, 102.9°E	23.0°/ -4.5°	/7	10.0°	53.5°N, 241.8°E	8.5°	N,R	Huang et al. (1986)
5#	Zhaojue	28.1°N, 102.9°E	18.1°/ -4.9°	/19	7.0°	54.8°N, 250.3°E	4.9°*	N	Liu et al. (1987)
6#	Yanyuan	27.6°N, 101.9°E	7.8°/ -3.9°	/17	10.7°	59.5°N, 266.4°E	7.6°*	N	Liu et al. (1987)
7	Meigu	28.3°N, 103.0°E	17.9°/ -5.8°†	/20	15.9°†	54.4°N, 251.2°E†	11.3°*	N	Zhou et al. (1988)
8	Miyi	26.5°N, 101.8°E	56.5°/-12.2°†	/19	8.8°†	26.3°N, 214.2°E†	6.4°*	N	Zhou et al. (1988)
9	Miyi	26.8°N, 101.8°E	234.5°/ 22.9°	/12	9.0°	24.3°N, 220.6°E	7.0°*	R	Zhou et al. (1988)
10	Dongchuan	26.1°N, 103.1°E	30.8°/ 5.7°	/8	25.0°	52.4°N, 225.9°E	16.5°	N,R	Huang et al. (1986)
11	Binchuan	25.6°N, 100.6°E	242.0°/ 1.3°	3/9	24.8°	24.7°N, 204.3°E	18.1°	R	Huang et al. (1986)
12	Kunming	25.0°N, 102.7°E	25.5°/-11.9°	10/57	6.1°	50.3°N, 241.0°E	4.5°*	N	Van der Voo et al. (1993)
13	Kunming	25.0°N, 102.7°E	244.1°/ 2.6°	1/2	-	22.7°N, 205.8°E	-	R	Van der Voo et al. (1993)
14	Zhijin	26.4°N, 105.7°E	224.5°/ 30.3°	1/5	12.0°	29.3°N, 235.3°E	9.9°*	R	Lin (1984)
15	Weining	26.7°N, 104.0°E	359.3°/ -5.9°	1/9	10.0°	60.3°N, 285.4°E	7.1°*	N	This study

Nr.: Numbers denoting directions in Fig. 4. N (n): Numer of sites (samples). α_{95} (A95): 95% confidence circles around mean directions (paleopoles). P: Polarity. ¶ from Zhang (1984). * estimated from d*m* and d*p*. # Directions have been checked with the original authors and paleopoles recalculated accordingly. † recalculated from original data subsets. Note that some duplicated results from Mount Emei are not included.

applying Watson's randomness test (1956), the vector sum of these ten samples is 2.65, much smaller than the critical value R_0 = 5.03 at 95% significance level, indicating that they are indeed from a statistically random population. This implies that the ChRM of the basalt clasts and thus the basalt underlying them was acquired before the formation of the pyroclastic breccia bed that contains them.

Discussion

Paleomagnetic results so far reported from the Emeishan Basalt are summarized in Table 1 and presented in Fig. 4. Various possibilities for explaining the difference within the predominant north-northeasterly group of directions derived from most studies and non-antipodality between this group of directions and the southwesterly down directions have been proposed by a number of workers (McElhinny et al., 1981; Lin, 1984; McElhinny, 1985; Huang et al., 1986; Zhou et al., 1986; Zhao and Coe, 1987), and were recently more thoroughly discussed by Enkin et al. (1992) and Van der Voo et al.(1993). Among these possibilities, the most discussed include (1) tectonic rotations of sampled localities: (2) remagnetization at much later times; (3) error in determination of bedding attitude and/or sample orientation; and (4) unaveraged secular variation. Since the sampled localities are spread over hundreds of kilometers away from the type section at Mount Emei which is close to the Longmenshan thrust fault and yield similar directions, and also because a general consensus has been reached that the northeasterly and shallow upward directions represent normal polarity, the possibility of large azimuthal rotations of more than 100° either on a regional or local scale (McElhinny et al., 1981; McElhinny, 1985) is no longer supported. However, Enkin et al. (1992) noticed that the paleomagnetic poles derived from individual studies were streaked along a small circle and suggested that minor rotations about local vertical axes between different localities might be responsible. We believe that this possibility can not be ruled out and might be a contributing factor to the difference, at least, in the mean ChRM directions of the same polarity observed at different localities. The mean direction for the basalt we calculated in this study is a little more northerly compared with other reported (north-)northeasterly ChRM directions, but it is certainly grossly similar to them, considering it is only based upon nine samples from a single flow and that secular variation may not be averaged out. The positive conglomerate test achieved on the pyroclastic breccia above the basalts effectively refutes the suspicion that the north-northeasterly ChRM directions are a remagnetization as late as Jurassic or Triassic (Lin, 1984; Van der Voo et al., 1993). Van der Voo et al. (1993) examined ferromagnetic minerals in eight basalt samples from Kunming with scanning electron microscopy (SEM). They found that titaniferrous magnetite is commonly cruciform in crystal habit in two samples that have NE and WSW declinations whereas there is pervasive alteration of titanomagnetite into titanium-free magnetite and sphene in six samples with NNE declinations. They suggested that the NE-SW directions are more likely to represent the Late Permian paleomagnetic field whereas the NNE directions might be a later remagnetization. The first author of this paper in the past made a systematic optical microscope inspection of thin sections of the Emeishan Basalt from most of the sampled localities and noticed pervasive alteration of major rock-forming minerals in the basaltic rocks from Yanyuan and Miyi of Sichuan Province and Bingchuan of Yunnan Province (Nrs. 6, 8, 9, 11 in Table 1). For instance, plagioclase was commonly replaced by albite, sericite, and clay minerals, and clinopyroxene (mainly augite) was often altered into actinolite, chlorite, and sometimes epidote. Although it is not impossible that the major rock-forming minerals such as plagioclase and clinopyroxene and opaque minerals like magnetite might be altered in different episodes of hydrothermal

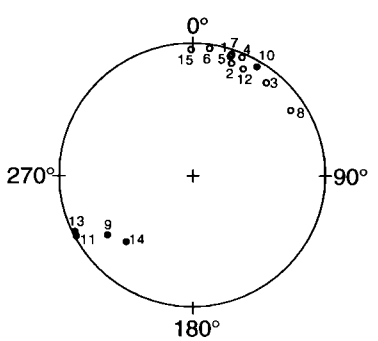


Figure 4. Equal-area projection of the mean directions observed from all the studies of the Emeishan Basalt. Numbers next to the circles are the same as in Table 1.

activity, it is unlikely that a hydrothermal fluid would pervasively alter plagioclase and clinopyroxene without affecting magnetite. The paleomagnetic samples from Miyi and Bingchuan happen to yield NE-WSW declinations (see Table 1). Therefore, we feel that SEM examination of only eight polished thin sections from one locality (Kunming) is insufficient to demonstrate an association of magnetic mineralogy with paleomagnetic directions. Since both solar and magnetic compass was used in some of the studies and no major discrepancy reported between these two kinds of orientation methods, systematic errors introduced by use of magnetic compass are unlikely. However, it is a well known fact that it is often difficult to accurately determine bedding attitude in massive basalt flows. Therefore, inaccuracy in determination of the bedding dip and strike is a possible factor that might contribute to the minor difference between the reported paleomagnetic directions. Also, it has to be realized that many of the published studies are of preliminary nature with usable samples less than twenty. In our view, only the sampling conducted by McElhinny et al. (1981), Liu et al. (1987, Nr. 3 in Table 1), Zhao and Coe (1989) and Van der Voo et al. (1993) are extensive enough to adequately average out secular variation of the geomagnetic field. This is probably especially true for the reversed southwesterly directions because so far only a few samples have been determined to be of reversed polarity in some of these studies. It is more likely that they represent spot readings of the Earth's magnetic field during a reversed polarity period. As a consequence, the normal and reversed directions would not be statistically antipodal.

In conclusion, the positive conglomerate test on the pyroclastic breccia supports the contention that the characteristic north-northeasterly and shallow up directions observed from the Emeishan Basalt represent a magnetization acquired at the time of formation of these rocks. The differences within the northeasterly group of directions as well as the non-antipodality between this group of directions and the reversed southwesterly down direction is likely to be due to a combination of factors such as minor rotations between sampled localities, inaccuracy in determination of bedding attitude, and inadequate sampling to average out the secular variation. To finally settle the question, more extensive investigations on carefully chosen sections, preferably using a magnetostratigraphic approach, are needed.

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